

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

ANSWERS

BIOLOGY
CLASS

2bi2____ / 2IPbi2__

REGISTRATION NUMBER

BIOLOGY

Paper 2

9648/02

26 August 2016

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and Biology class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Sections A - D

Answer **all** questions in the spaces provided on the question paper.

Section E

Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	(Total: 23)
1	/ 12
2	/ 11
Section B	(Total: 19)
3	/ 7
4	/ 12
Section C	(Total: 19)
5	/ 12
6	/ 7
Section D	(Total: 19)
7	/ 7
8	/ 12
Section E	(Total: 20)
9 or 10	/ 20
Total	/ 100

This document consists of **28** printed pages.

Section A

Answer **all** the questions in this section.

- 1** Many bacteria can digest cellulose using a group of enzymes called cellulases. Cellulases **A** and **T** were extracted from two different bacteria, *Agrobacterium tumefaciens* and *Thermotoga maritima*, respectively.

Fig. 1.1 shows the results of an investigation into the effect of temperature on the activity of each enzyme.

L represents the lowest temperature at which activity of each enzyme was detected.
H represents the highest temperature at which activity of each enzyme was detected.

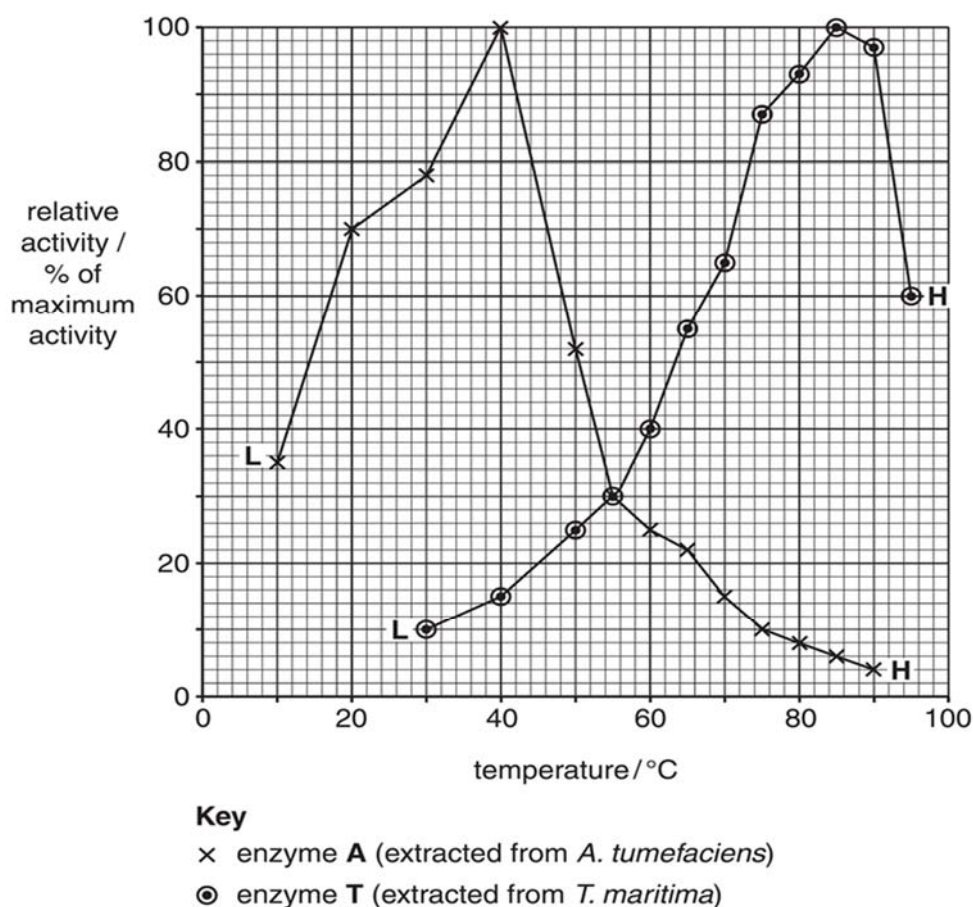


Fig.1.1

- (a) With reference to Fig. 1.1, describe two differences in the results for the two enzymes, **A** and **T**. [2]

- 1. Optimum temperature for enzyme A (40°C) is lower than that for enzyme T (85°C).**
 (Accept: "enzyme A optimum temperature is lower than enzyme T by 45 °C")
 (Accept: "maximum activity of enzyme A is at a lower temperature than T")
- 2. Temperature range for enzyme A (10–90°C / spans 80°C range) is greater than that for enzyme T (30–95°C / spans 65°C range).**

[2]

3. Difference in shape of curve before or after optimum

E.g. before optimum, enzyme A has a steep increase, whereas enzyme T has a more gradual increase.

E.g. after optimum, enzyme T has a steep decrease, whereas enzyme A has an initial steep decrease followed by a subsequent less steep gradient (Accept: enzyme A has a gradual decrease).

4. Lowest temperature for enzyme A (at 10°C) as compared to lowest temperature for enzyme T (at 30°C)

5. Highest temperature for enzyme A (at 90°C) as compared to highest temperature for enzyme T (at 95°C)

6. Enzyme A works better at lower temperatures (10-55°C) as compared to enzyme T, which works better at higher temperatures (55-95°C).

@ 1mark each, max 2

- (b)** With reference to Fig. 1.1, explain the effect of increasing temperature on the relative activity of enzyme T.

To explain the effect,

- 1. As temperature increases from 30°C to 95°C, increase in kinetic energy of enzyme and substrate molecules, hence increasing the rate of effective collisions between them.**
- 2. This facilitates the formation of more enzyme-substrate (ES) complexes resulting in an increase in relative activity.**
- 3. However, as the temperature increases beyond the optimum temperature of 85°C, the enzyme molecules start to vibrate violently with the excess kinetic energy. As a result, the weak (non-covalent) intramolecular bonds break. These bonds maintain the secondary and tertiary structures/ specific 3D conformation of the enzyme. Enzyme becomes denatured.**

[3]

- (c)** Suggest a structural feature of enzyme T, which helps to explain the results obtained in the investigation.

- 1. Enzyme T contains many cysteine amino acid residues (that maintain the tertiary structure by forming disulphide bonds).**
- 2. Higher temperature required to break the many/multiple disulphide bonds within enzyme T (cysteine R-groups form covalent disulphide bonds)**

[2]

Fig. 1.2 shows the structure of small sections of DNA and messenger RNA (mRNA) in the nucleus of *Agrobacterium tumefaciens* during transcription of the gene coding for cellulase.

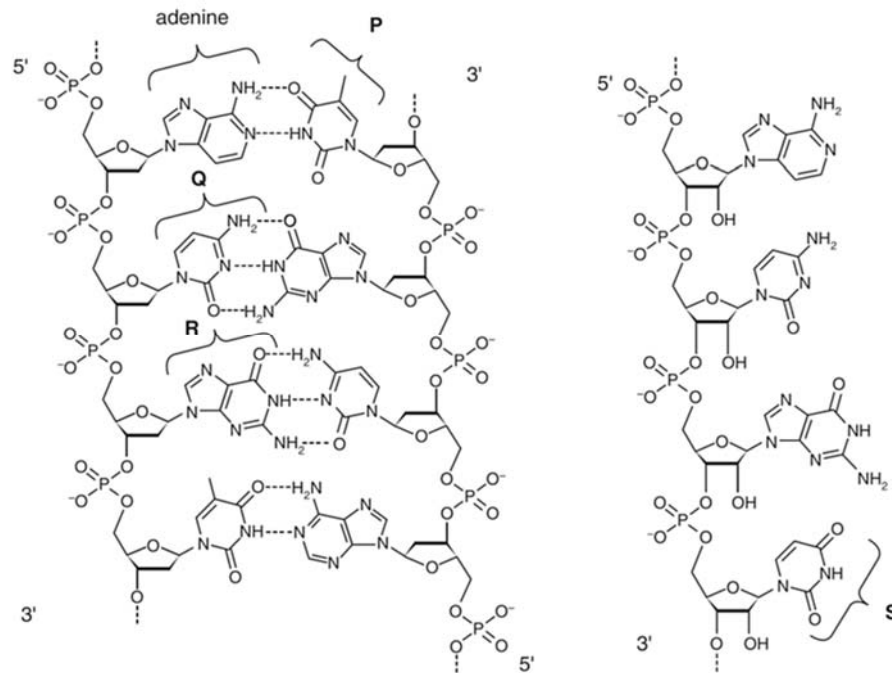


Fig. 1.2

(d) Name the bases P to S.

P: Thymine , Q: Cytosine , R: Guanine , S: Uracil

(all 4 correct – 2 mark, 2-3 correct – 1 mark)

[2]

(e) Describe how messenger RNA coding for cellulase is synthesised in *Agrobacterium tumefaciens*.

1. RNA polymerase (with the aid of Sigma factor) binds to cellulase gene promoter (Pribnow box), causing the DNA double helix to unzip (separate) [3]
2. One of the two DNA strands serves as template strand
3. Free ribonucleotides / free ribonucleoside triphosphates / rNTPs form complementary base pairing with bases/deoxyribonucleotides on the template DNA strand
4. RNA polymerase catalyses formation of phosphodiester bonds (between ribonucleotides via condensation reactions)
5. Template DNA strand is read from 3' to 5' direction, and mRNA is synthesized from 5' to 3' direction
6. Transcription ends when RNA polymerase transcribes a terminator sequence

@1 mark each, max 3

[Total: 12]

- 2 Fig. 2.1 shows the main steps involved in the synthesis of preproinsulin to insulin in the pancreatic β -cell. The preproinsulin is synthesised into the lumen of organelle **A** as proinsulin.

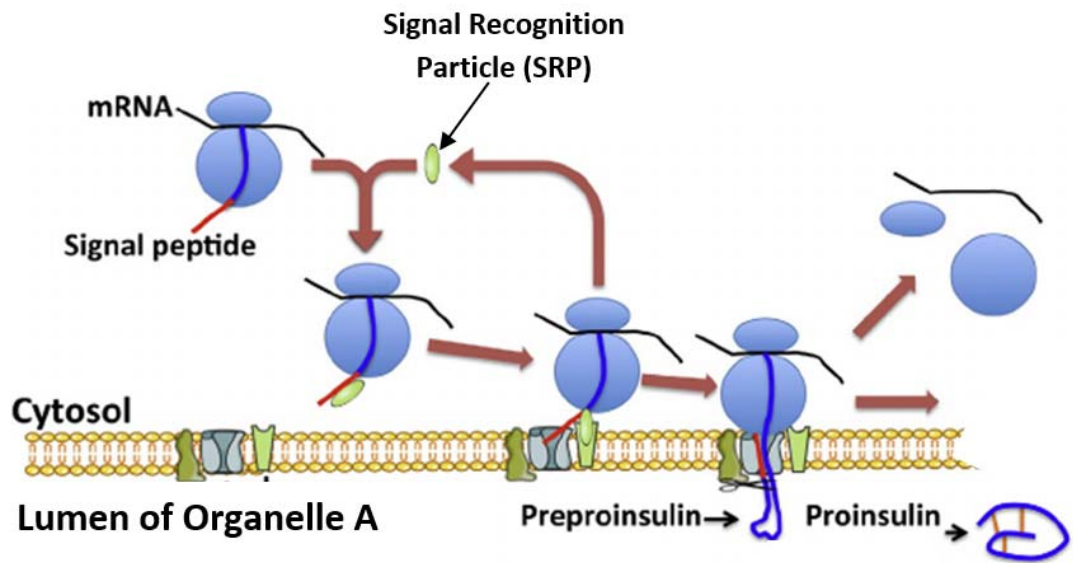


Fig. 2.1

The proinsulin is then be transported to organelle **B** where it is further processed to form insulin.

Fig. 2.2 shows the conversion of proinsulin to insulin in organelle **B**.

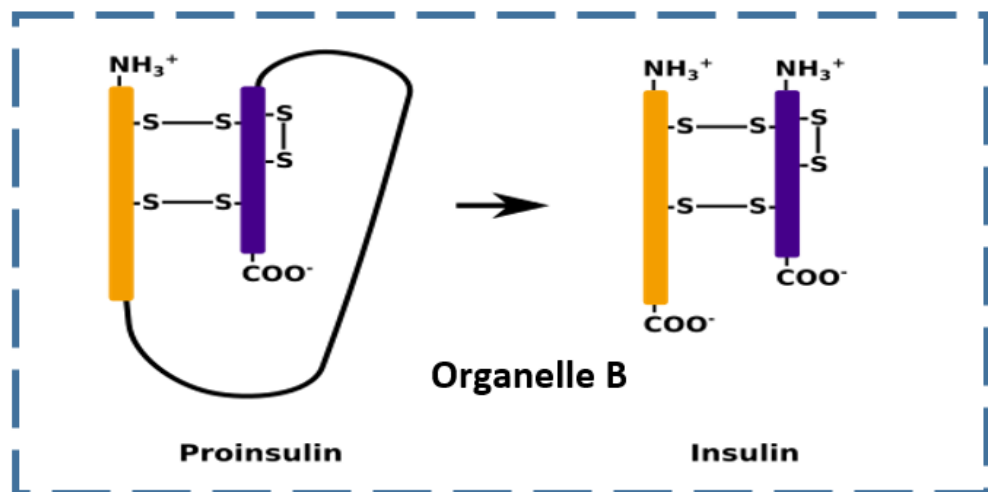


Fig. 2.2

- (a) Name the organelles labelled **A** and **B**.

Organelle A: Rough Endoplasmic reticulum (Must spell in FULL)

Organelle B: Golgi apparatus / Golgi body

[1]

(b) State the role of rRNA in insulin synthesis.

1. rRNA along with ribosomal proteins forms the structural component of ribosome (large and small sub-unit).
2. rRNA is responsible for catalytic function of ribosome in the formation of peptide bond between amino acids (found at the large sub-unit).
3. rRNA in the small ribosomal subunit binds to 5' end of mRNA sequence during protein translation.
4. rRNA at the A site binds to the amino-acyl tRNA while the rRNA at the P site binds to the peptidyl-tRNA.

OWTTE

* Any 2 of the above

[2]

(c) Insulin is released by pancreatic β -cell. Outline the route taken by proinsulin.

1. From Rough ER, a transport vesicle takes the proinsulin to the Golgi apparatus (GA).
2. After chemical modification and packaging, a secretory vesicle pinched/buds off from GA.
3. The transport of secretory vesicles (aided by microtubules-the cytoskeletal elements) in the cytoplasm, until they fuse to the plasma membrane.
4. Secretory vesicle fuses with the cell surface membrane before releasing insulin by exocytosis.

[2]

* 3 points to get 2 marks. Pt 4 is crucial to talk about.

The synthesis of insulin is regulated at the transcriptional levels in β -cell.

Fig. 2.3 shows the binding of three β -cell-specific transcriptional factors, Pdx-1, MafA and NeuroD1 E47 in response to high glucose levels in β -cell.

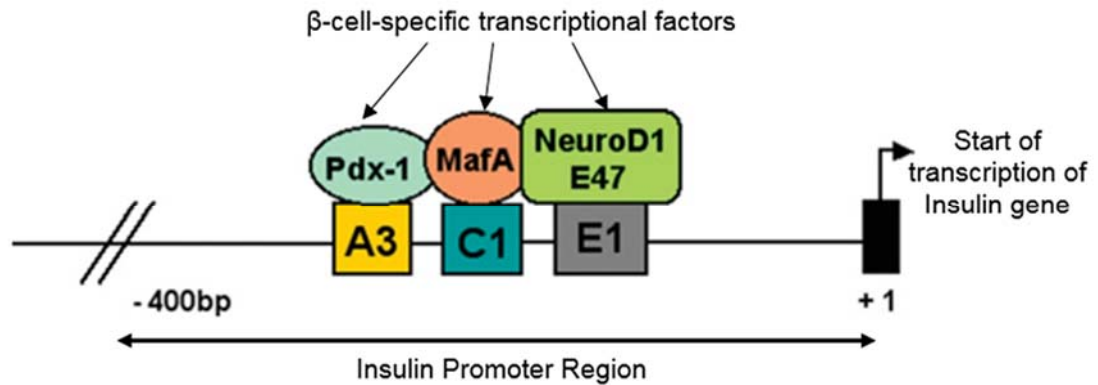


Fig. 2.3

(d) Explain how β -cell-specific transcriptional factors will lead to high level of insulin gene expression.

[2]

1. β -cell-specific transcriptional factors are activators that bind to the enhancer region found in the insulin promoter region.
2. Binding of β -cell-specific transcription factors will lead to increase in the formation of transcription initiation complex / helps RNA polymerase, transcription initiation factors to bind to promoter region with greater affinity.
3. β -cell-specific transcriptional factors act as activators and lead to chromatin remodelling / changes in chromatin structure to increase rate of transcription. E.g. via histone acetylation to allow greater accessibility to RNA polymerases for transcription to occur.

E.g of changes:

- a. Acetylation of histone tails / addition of acetyl groups catalyzed by histone acetyltransferase
- b. Cause, the chromatin structure becomes less compact and allow access to RNA polymerase to insulin gene promoter sequence. Transcription can take place.

Any 2 of the above point

- (e) It has been found that insulin promoters differ widely in efficiency among individuals. Strong promoters cause frequent initiations of transcription while weak promoters have low frequency of initiations of transcription. Explain what may have caused the difference in efficiency of the insulin promoters.

1. **Gene mutation / base substitutions in the insulin promoter sequence result in changes in the base sequence of the promoter**
2. **This change will cause the structure of the promoter to be of less complementary to the DNA binding site of the RNA polymerase and affects the binding of the RNA polymerase to the promoter sequence / Resulting in a promoter which initiates transcription less strongly;**

Other accepted answers:

- **Promoter region found in tightly coiled region of DNA molecule, affects accessibility by RNA polymerases- Level of supercoiling of the DNA affects accessibility to RNA polymerase.** [2]
 - **Methylation of the promoter regions, which affects the binding by RNA polymerases.**
- (f) Explain why gene mutations do not always produce mutated insulin protein whereas mutations of the splicing sites involved in RNA splicing will produce mutated insulin. [2]

Why gene mutations do not always produce mutated insulin:

1. **Gene mutations that involve substitution may result in the same amino acid being coded for and due to the degenerate code/ same amino acid can be coded for by different codons.**
2. **Gene mutation could occur at the intro region instead of exons.**

Why mutations at RNA splicing sites will produce mutated insulin:

Mutation at the splice site will affect the binding of spliceosome, and will affect the removal of introns & exons, hence giving rise to a mutated protein with loss of function.

Example of the kind of mutations at RNA splicing sites and the outcome:

An example of mutation at RNA splicing sites (<i>any one</i>)	Effect of such mutation (i.e. production of mutated collagen) (<i>any one</i>)
Different combinations of exons being produced	Different primary sequences of amino acids resulting in different protein (mutated protein)
An exon is lost/ wrong excision of exons	Large number of bases and hence amino acids is lost/ as above
Introns not removed by spliceosome	Introns translated and became additional amino acids, this will lead to change the protein structure

[Total: 11]

Name: _____ Class: 2bi2___ / 2IPbi2___

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Section B

Answer **all** the questions in this section.

- 3 Fig. 3.1 shows the life cycle of a water flea.

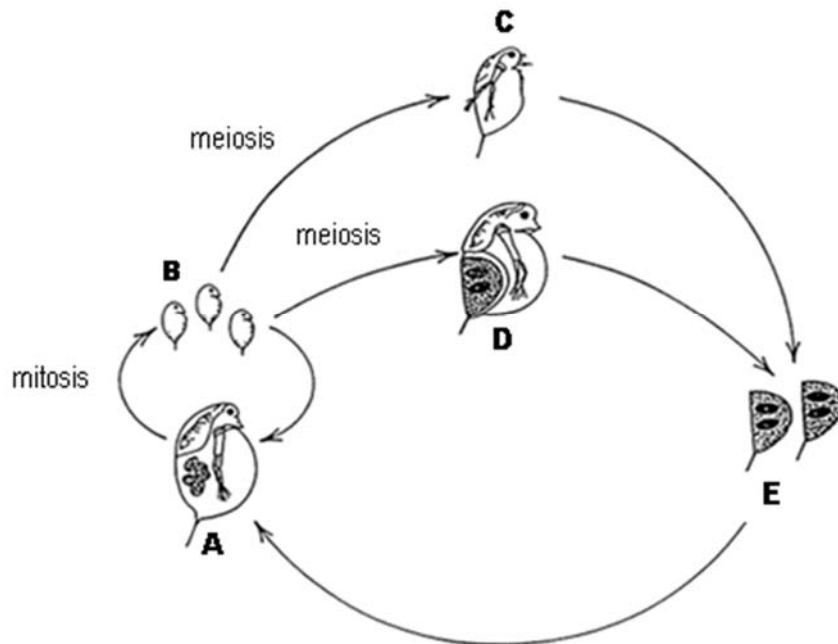


Fig. 3.1

In favourable conditions, all the animals in a population are females (**A**). These females produce eggs by mitosis, which develop into young females (**B**) without being fertilized. In unfavourable conditions, eggs produced by meiosis develop directly without fertilization into either males (**C**) or females (**D**). The eggs produced by the females (**D**) are fertilized by the sperms from the males (**C**), then released in protective egg cases (**E**) which enable them to survive unfavourable conditions. When favourable conditions return, these eggs develop back into females (**A**).

- (a) The females at stage **A** of the life cycle have 18 chromosomes.

Complete the table to show the number of chromosomes at the other stages of the life cycle.

stage of life cycle	chromosome number
A	18
B	18
C	9
D	9
E	18

[1]

(b) Explain why the eggs from **D** and the sperms from **C** must be produced by mitosis.

1. Since **C** and **D** (developed from unfertilised eggs from **B**) are haploid, mitosis ensures that the haploid chromosome number is preserved / the eggs and sperms are haploid.
2. Thus, when the haploid sperm and haploid egg fuse, the original diploid chromosome number is restored.

[2]

(c) Explain why females **A**, developed from fertilized eggs **E**, are genetically different from each other.

(Any 3)

1. **C** and **D** developed from eggs that are produced by meiosis in **B**.
2. Crossing over between non-sister chromatids of homologous chromosomes at prophase 1 of meiosis
3. Independent assortment of homologous chromosomes at metaphase 1 of meiosis
4. Independent assortment of non-identical chromatids at metaphase 2 of meiosis
5. Random mating between **C** and **D**
6. Mutations can occur at any time.

[3]

(d) Give an example of a favourable condition in which females will develop from eggs formed via mitosis.

(Any 1)

- Presence of water in a previously dry pond
- Reasonably high temperature (~20°C)
- Abundant food source
- Lack of competition
- Stable environment
- Few or no predators
- Appropriate photoperiod
- Water of optimal pH
- Suitable salinity
- (any other valid point)

[1]

[Total: 7]

- 4 Cattle were found to have three different coat colours: brown, white and roan. Roan cows have both brown and white patches. When two roan cattle were crossed, a ratio of 1:2:1 was obtained for brown, roan and white coat colour respectively. A separate gene on X chromosome in cattle code for a disease called Agnathia. The absence of a normal allele causes a deformity in the lower jaw.

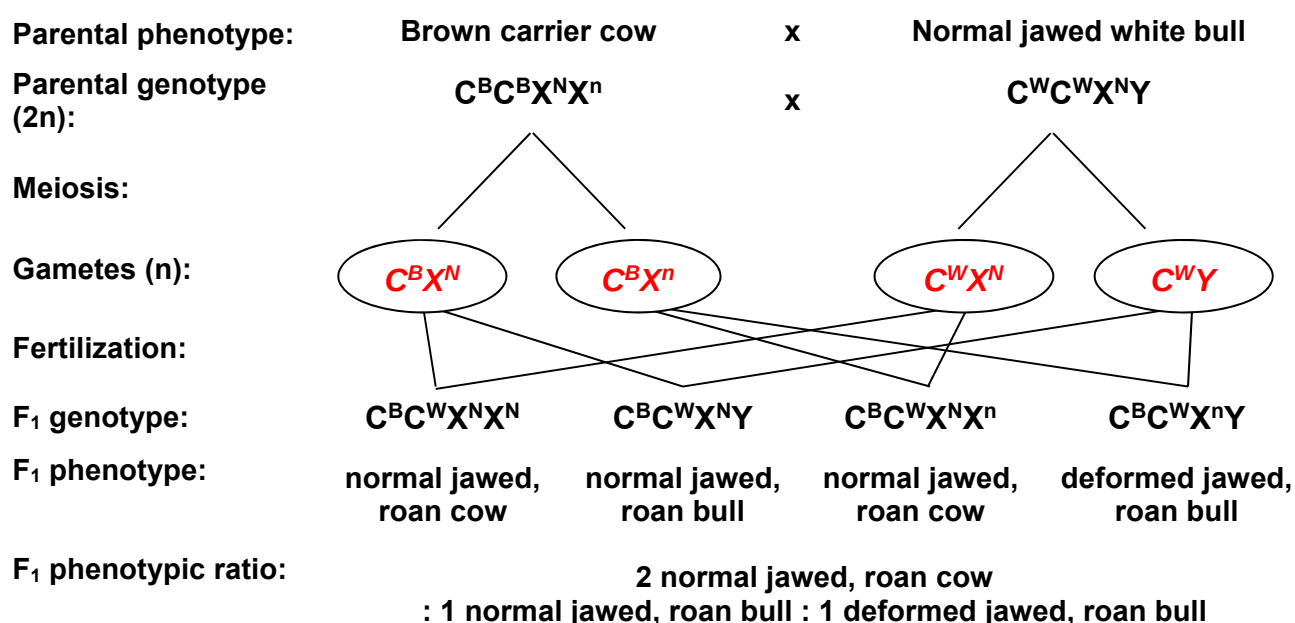
- (a) Construct a genetic diagram to show the expected ratio when brown carrier cows are crossed with normal jawed white bulls.

Let C^B represent the codominant allele for brown coat.

Let C^W represent the codominant allele for white coat.

Let X^N represent the X-linked dominant allele for normal jaw.

Let X^n represent the X-linked recessive allele for deformed jaw.



Both parental phenotypes and genotypes are correct (with appropriate use of symbols). [1]

All possible gametes from each parent are correct. [1]

Genetic diagram correctly shows 4 possible combinations of gametes. [1]

The phenotypes of all F₁ genotypes are correct. [1]

Expected F₁ phenotypic ratio is correct. [1]

[5]

(b) State the mode of inheritance of the two traits. Explain how you arrive at your answer.

- **agnathia: X-linked / sex-linked recessive**
- **gene is on X chromosome AND absence of normal allele causes it**
- **coat colour: codominance**
- **roan cattle have both brown and white patches (intermediate phenotype)**

Award 1 m if mention any 2 of the above

Award 2 m if mention all of the above

[2]

(c) When a geneticist carried out a cross between brown carrier cows and normal jawed white bulls, he obtained the following phenotypes.

normal jawed, roan cows	55
normal jawed, roan bulls	37
deformed jawed, roan bulls	28

Fig. 4.1 shows the table of probabilities.

df	probability				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.69	13.28	18.47
5	9.24	11.07	13.39	15.09	20.52

Fig. 4.1

The formula for the chi-square statistic used in the chi square test is as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \text{where } \begin{array}{l} O = \text{observed value;} \\ E = \text{expected value.} \end{array}$$

[3]

With reference to Fig. 4.1, carry out a chi-square test to support your explanation in (b). Show your workings clearly in the space below.

Null Hypothesis: H_0 : There is no significant difference between observed and expected data. i.e. the observed data follows a 2:1:1 distribution. Any difference is due to chance.

Phenotype	Ratio	(O)	(E)	$(O-E)^2/E$
Normal jawed, roan cows	2	55	60	0.42
Normal jawed, roan bulls	1	37	30	1.63
Deformed jawed, roan bulls	1	28	30;	0.13;
		$\Sigma(O) = 120$	$\Sigma(E) = 120$	$X^2_{cal} = 2.18;$

Correct working for calculation of chi-square statistic (i.e. 2.18) [1]

Use correct df (i.e. 2) and probability (i.e. 0.05) in Fig. 4.1 to determine critical chi-square value (i.e. 5.99) [1]

State / Accept the null hypothesis [1]

(d) In some cases, the environment may affect the phenotype.

Give one named example to illustrate such an environmental effect.

[2]

- (valid character) e.g. fur colour / synthesis of black pigment
- (type of organism) e.g. Himalayan rabbit
- (change in environment) e.g. exposure to heat
- (corresponding change in phenotype) e.g. white fur formed instead of black fur / no synthesis of black pigment

[Total: 12]

Section CAnswer **all** the questions in this section.

- 5** Hepatitis C virus (HCV) is an enveloped, positive-strand RNA virus within the family Flaviviridae. HCV undergoes reproductive cycles similar to other enveloped viruses, such as influenza virus and dengue virus.

Fig. 5.1 shows a model of a HCV particle.

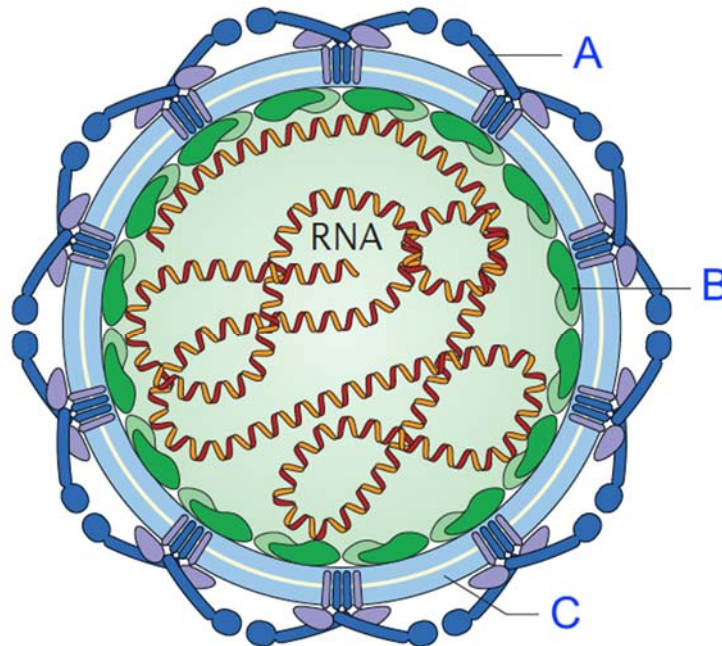


Fig. 5.1

- (a)** Identify the labelled structures **A**, **B** and **C**.

A: glycoprotein/ envelope glycoprotein

B: capsid/ capsomere

C: envelope

[2]

- (b)** Explain why HCV is considered an obligate parasite.

1. HCV can only survive and reproduce in living host cells.

2. HCV uses host cell's machinery and resources to replicate and assemble new viral particles.

[2]

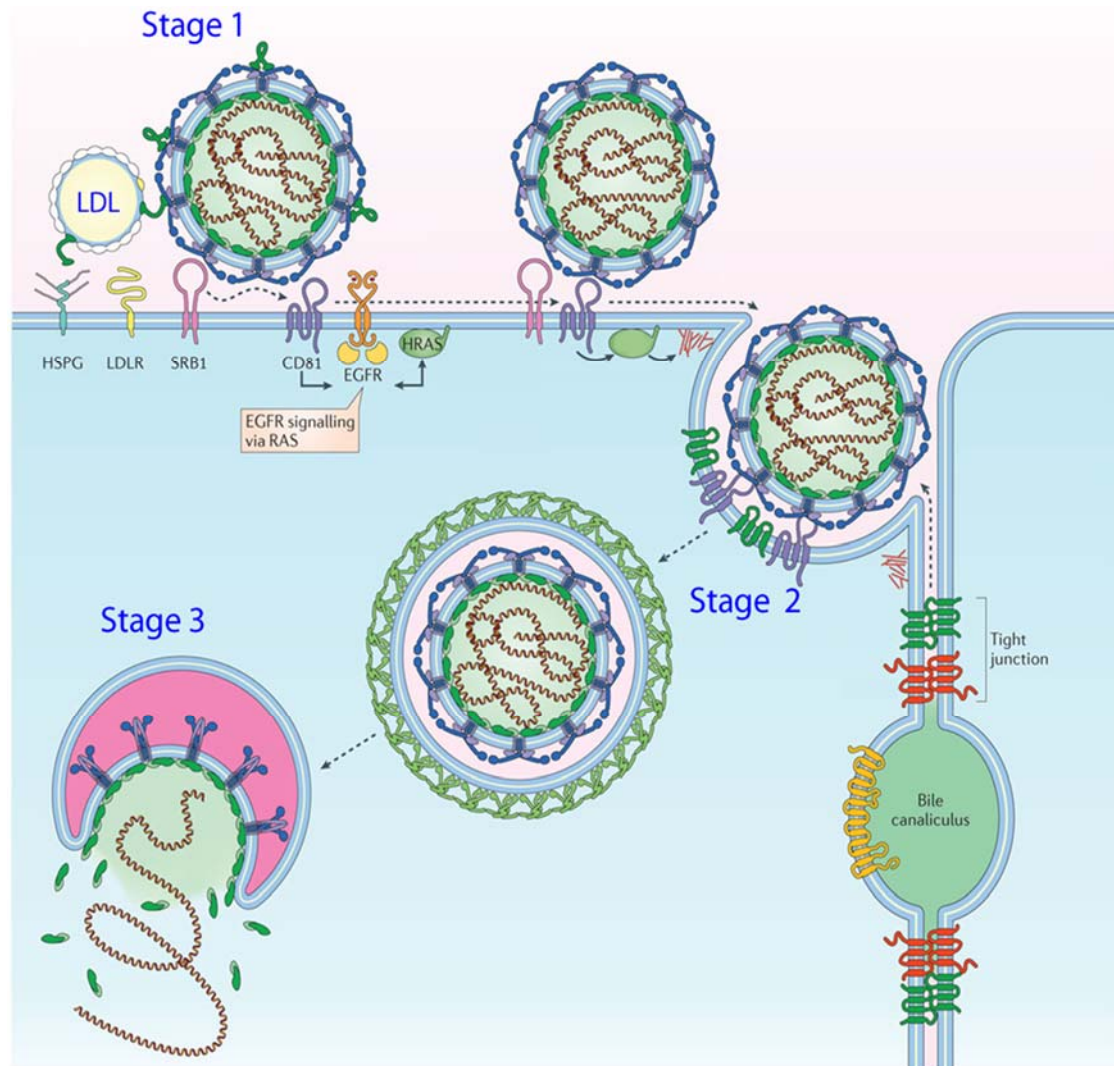


Fig. 5.2

(c) With reference to all labelled stages in Fig. 5.2, describe how HCV gains entry into its host cell.

1. HCV binds to SRB1/ CD81 in stage 1.
2. Host cell membrane invaginates and HCV enters host cell through receptor-mediated endocytosis in stage 2.
3. Low pH within endosome stimulates fusion of viral envelope with endosome membrane, exposing capsid to digestion by cellular enzymes, releasing viral RNA into the cytoplasm in stage 3.

[3]

Fig. 5.3 shows the positive-strand RNA genome of HCV genome.

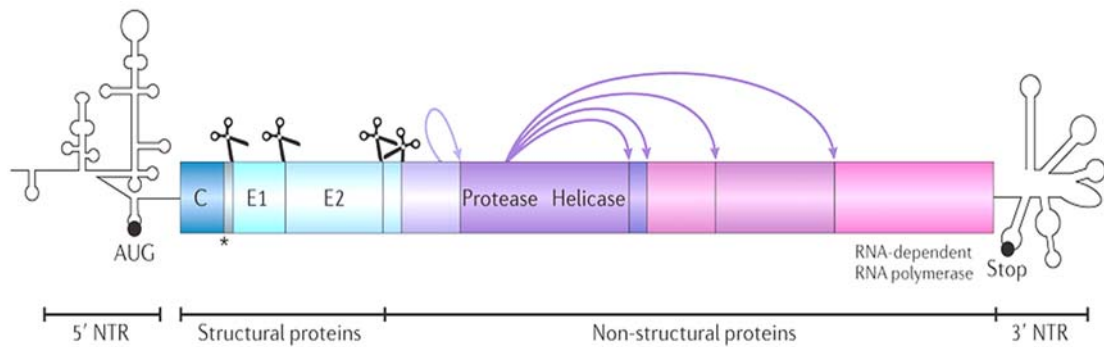


Fig. 5.3

(d) With reference to Fig. 5.3, describe the role of HCV genome in the infection process.

1. **Positive-stranded RNA is translated into a polyprotein by host ribosomes and cleaved into individual proteins.**
2. **RNA-dependent RNA polymerase coded by viral genome synthesizes a negative-sense RNA intermediate which is used as a template for synthesizing**

[2]

Hepatitis B virus (HBV) and HCV are known to be two major causative agents of hepatocellular carcinoma. Studies have shown that HBV DNA can be integrated during the early stages of infection. The integration of viral DNA is associated with deletions in portions of the host chromosomes. Many of these chromosomal segments contain known genes such as p53.

(e) Explain how HBV infection may lead to cancer.

1. **p53 is a tumour suppressor gene, which bring about inhibition of cell cycle and activation of DNA repair genes when DNA damage is detected.**
2. **Deletion of p53 results in loss of function mutation of p53, allowing DNA damage to accumulate in cells.**
3. **As cells accumulate 4-6/ more mutations in key regulatory genes, including appearance of at least 1 active oncogene, they become cancerous.**

[3]

[Total: 12]

- 6 (a) Human newborns and hibernating mammals contain large amounts of brown adipose tissue ('body fat').

Fig 6.1 shows the electron micrograph of a brown adipocyte. Brown adipocytes are characterised by the presence of numerous vacuoles and organelle X throughout the cell.

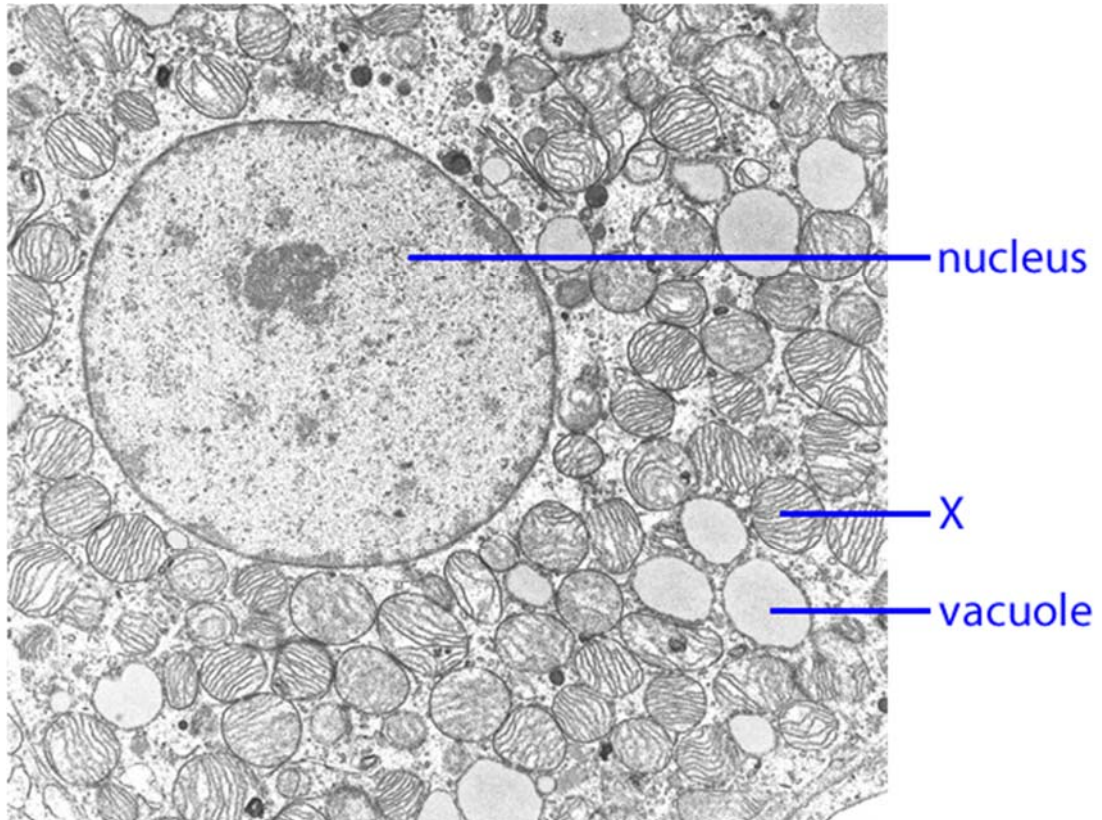


Fig. 6.1

- (i) Identify organelle X.

mitochondrion

[1]

- (ii) Suggest the role of the numerous vacuoles found in brown adipocytes.

They store lipids / triglycerides / fats.

[1]

- (b) Fig. 6.2 shows the schematic representation of a series of protein complexes found on the inner membrane of organelle X.

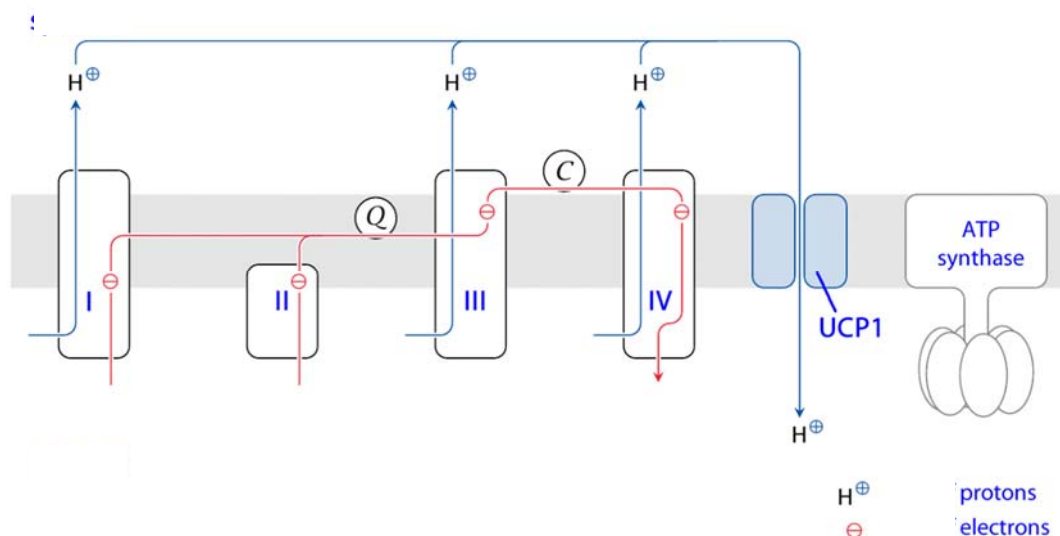


Fig. 6.2

- (i) Oxygen is required to sustain the process illustrated in Fig. 6.2. With reference to Fig. 6.2, describe the role played by oxygen.

Oxygen serves as the final electron acceptor, receiving electrons from complex IV to form water.

[1]

- (ii) Brown adipocytes contain a unique protein, UCP1, which is not found in organelle X in any other cell type.

Evaluate the impact of UCP1 on the normal functioning of the process illustrated in Fig. 6.2 and suggest the physiological significance of brown adipose tissue.

- As UCP1 allows protons to leak back into the matrix without passing through the ATP synthase, no ATP will be synthesized from the NADH and FADH₂.**
- The energy released from the spontaneous flow of protons through UCP1 is lost as heat, which helps to keep the organisms warm.**

[2]

- (c) In other cell types, NADH and FADH₂ are used to drive ATP synthesis by ATP synthase. Using relevant information from Fig. 6.2, suggest and explain why more ATP is produced from NADH.

- NADH and FADH₂ donates electrons to complex I and II respectively, the energy released from transfer of electrons through the complexes is used to pump protons across the inner membrane.**
- Because NADH started with Complex I, it had more chances to pumps more protons across the gradient, which powers the ATP synthase and gives us 3 ATP per molecule of NADH, while FADH₂ produces 2 ATP during the ETC because it gives up its electron to complex II, bypassing complex I.**

[2]

[Total: 7]

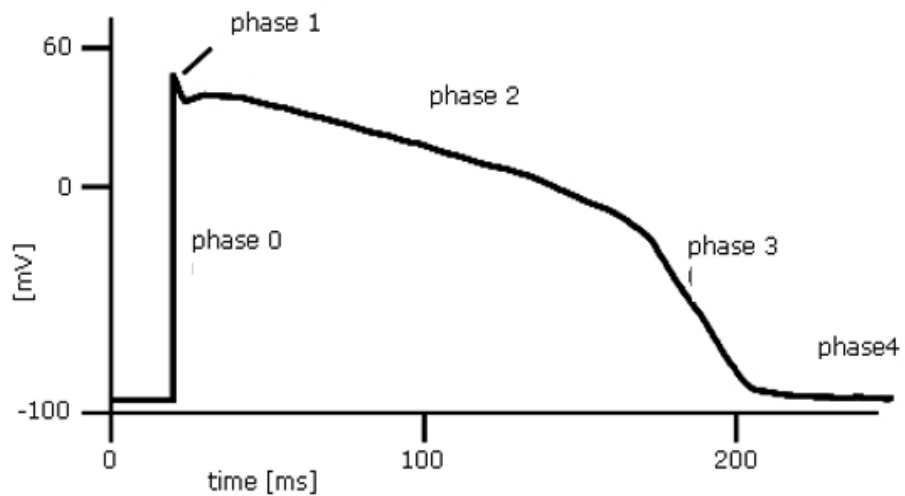
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19

Section DAnswer **all** the questions in this section.

- 7 The cardiac action potential is a specialised action potential with unique properties necessary for the electrical conduction system of the heart. The cardiac muscle cells share many things in common with nerve cells.

Fig. 7.1 shows the five phases of a cardiac action potential. Each of which is characterised with their respective changes in membrane potential.

**Fig. 7.1**

The table shows the relative concentrations of ions inside and outside of the cardiac muscle cells.

element	ion	extracellular	intracellular
sodium	Na ⁺	135 – 145	10
potassium	K ⁺	3.5 – 5.0	155
chloride	Cl ⁻	95 – 110	20 – 30
calcium	Ca ²⁺	2	10 ⁻⁴

- (a) Estimate the resting potential for the cardiac action potential shown in Fig. 7.1.

Between (-90 mV to -95 mV)

[1]

(b) Using the information from the table, explain how the resting potential can be produced in the heart muscles using sodium and potassium ions.

- **[Na⁺] outside the cell > [Na⁺] inside the cell while [K⁺] outside the cell < [K⁺] inside the cell**
- **Possible involvement of a protein pump: Requires energy in the form of ATP for movement of ions against concentration gradient**
- **More Na⁺ pumped out than K⁺ pumped in → Nett loss of +ve ions from inside the cell thus inside –ve 90mv**

[3]

(c) Compare phase 2 of the action potential of a cardiac muscle cell with that of a nerve cell.

- **The fall in membrane potential in a nerve cell is more drastic/ shows a steeper gradient/ faster rate/ longer time in OR**
- **Nerve cell only shows Phase 3/ Phase 2 is absent in nerve cells.**

[1]

(d) Cardiac arrhythmia refers to any abnormal electrical activity in the heart. As a result, the heart may beat too fast. Calcium channel blockers such as Verapamil are often used to treat this condition.

Suggest and explain the action of Verapamil in controlling this symptom of arrhythmia.

[2]

- **Interrupts inflow of Ca²⁺ during plateau phase to prolong the refractory period/ depresses Phase 2 and 3**
- **Reduces the rate of heart contraction → heart rate declines**

[Total: 7]

- 8 The table shows the amino acid differences in the cytochrome b protein between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

Fig. 8.1 shows the phylogenetic tree based on differences between the cytochrome b proteins.

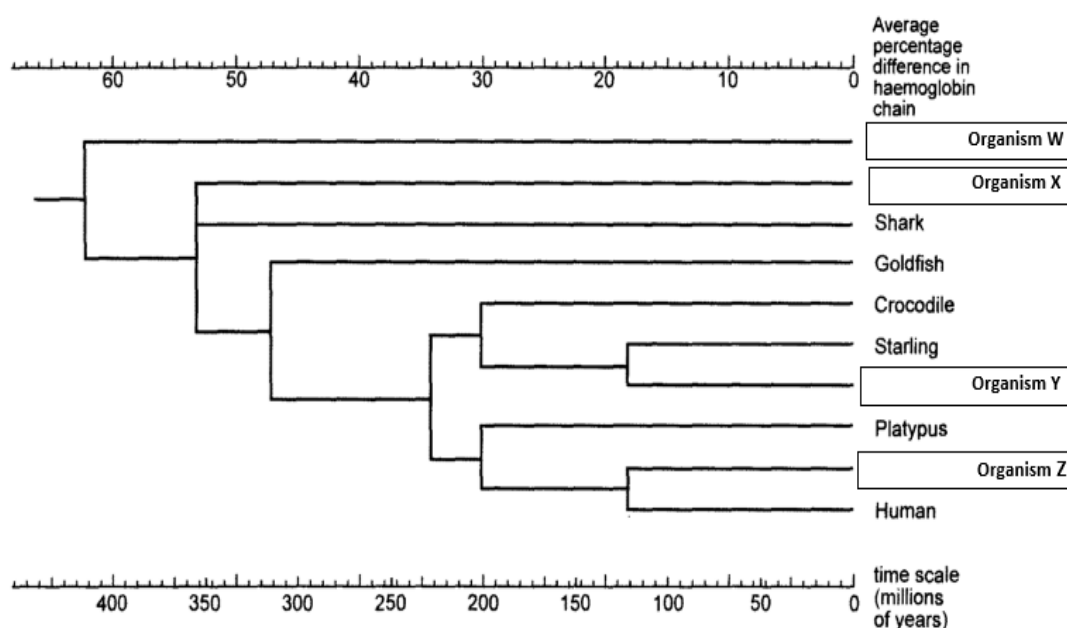


Fig. 8.1

- (a) Using information from the table and Fig. 8.1, identify organisms **W** to **Z**.

W: lungfish

X: coelacanth

Y: ostrich

Z: elephant

[2]

- (b) Explain how differences in amino acid sequences in the cytochrome b chain allow the establishment of the phylogenetic tree.
- **% of aa difference in Hb chain indicates relatedness;;**
 - **few difference indicates recent common ancestor / large difference indicates early divergence OWTTE**
 - **provides quantitative data to construct phylogenetic tree**
- [3]
- (c) Explain the difference between classification and phylogeny.
- **classification refers to grouping organisms based on similar characteristics**
 - **characteristics may be analogous and not homologous**
 - **phylogeny involves grouping organisms based on evolutionary relationship**
 - **similarity is due to inheritance from common ancestry**
- [2]
- (d) Suggest why homology still features prominently in evolutionary studies despite the advantages that molecular evidence can confer.
- **easier to use / often requires observation rather than machines**
 - **DNA / protein not always available e.g. in fossils**
- [2]
- (e) Explain the role of neutral mutations in evolutionary studies.
- **constant rate of mutation**
 - **act as molecular clock**
 - **genetic differences act as an indicator of time / period of divergence**
 - **genetic differences act as an indicator of speciation event OWTTE**
- [3]

[Total: 12]

- 9 (a) Compare the role of nervous system and endocrine system as communication systems within organisms. [6]

Similarities

- S1. Both systems communicate through use of signalling molecules that bind to specific receptors on effector cells.**
- S2. They function in response to stimuli**

Differences

- D1. Nervous system transmit information in the form of electrical impulses along an axon and in the form of chemical signals known as neurotransmitters across a synapse. Endocrine system transmit information in the form of chemical signals known as hormones only.**
- D2. Signals are transmitted via specific neural pathways consisting of communicating neurons in the nervous system. In contrast, signals are transported by the circulatory system via bloodstream.**
- D3. Transmission of signals via nervous system is rapid, where transmission of nerve impulses to effector cells bringing about response is completed in milliseconds. Transmission of signals via endocrine system is slow, which may take minutes to days for hormones to be produced and carried by blood to target organs for response to occur.**
- D4. Communication through nervous system results in localized responses from the target cell(s) that are post-synaptic to the motor neurons. In contrast, endocrine system results in responses that may be widespread as various tissues/ organs can respond to a single hormone.**
- D5. The response brought about by nervous system is immediate and short-lived, while the response brought about by endocrine system is slow and long-lasting.**
- D6. Communication via nervous system results in “all or none” response, where the magnitude of action potential is the same regardless of the strength of the stimuli. Whereas communication through endocrine system can bring about graded response.**
- D7. The control of response in nervous system may be voluntary or involuntary, while the control of response in endocrine system is always involuntary.**

- (b) Explain the meaning of the term homeostasis with specific reference to the control of raised blood glucose concentration in humans. [8]

1. Homeostasis is the maintenance of a constant internal environment.
2. One of the principles underpinning homeostasis is self-regulation, in that the control mechanism is triggered by the parameter that is being regulated.
3. Another principle of homeostasis is negative feedback, whereby deviation from set/reference point triggers response which counteracts/ reverses the deviation, restoring the parameter to set/reference point.
4. The set/reference point for blood glucose level is 90mg glucose/100cm³.
5. The rise in blood glucose level from set point is the stimulus that triggers the control mechanism to reduce blood glucose level back to set point.
6. The rise in blood glucose level is detected by the alpha and beta cells of the Islets of Langerhans in the pancreas (detector).
7. Beta cells are triggered to secrete insulin into bloodstream,
8. Alpha cells are signalled to stop glucagon secretion.
9. Insulin is transported via bloodstream to target cells, such as liver cells (effector). Insulin binds to receptors and acts to bring about responses that restore the set point.
10. One response is increased fusion of vesicles containing glucose-carrier/glucose-transporter proteins with the plasma membrane so as to increase rate of uptake of glucose in the effector cells.
11. These cellular responses result in a decrease in blood glucose level back to the set point of 90mg glucose/100cm³ of blood. This is negative feedback regulation of blood glucose level.
12. As blood glucose level return to the set point, beta cells will no longer be triggered to secrete insulin, hence insulin level in the blood decreases.

(c) Describe the cell signalling pathway that glucagon initiates in order to regulate blood glucose concentration. [6]

1. Glucagon binds to extracellular binding site of G protein-coupled receptor (GPCR) on the plasma membrane of liver cells.
2. Binding of glucagon triggers a change in 3D conformation in the GPCR, resulting in the release of GDP followed by binding of GTP, thus activating G protein.
3. (a) Activated G protein dissociates from receptor and diffuses along the plasma membrane to bind to and activate adenylyl cyclase.; (b) Adenylyl catalyses the conversion of many cyclic AMP (cAMP) from ATP.;
4. cAMP acts as second messenger to activate enzymes such as protein kinase A.
5. Protein kinase A activate other enzymes by phosphorylating them, triggering a phosphorylation cascade that help to amplify the initial signal and bring about the necessary cellular responses.
6. For example, glycogen phosphorylase brings about increased glycogenolysis by catalysing the breakdown of glycogen to glucose-1-phosphate which is then converted to glucose and transported out of the liver cell into the bloodstream to increase blood glucose level back to the set point.
7. Another cellular response is increased activity of enzymes involved in gluconeogenesis (synthesis of glucose from non-carbohydrate sources) forming more glucose to increase availability of glucose for cellular respiration

[Total: 20]

10 (a) Compare the structural and regulatory genes in prokaryotes.

[6]

(Maximum 2 marks for similarities)

- Both do not contain introns.
- Both are transcribed by RNA polymerase.
- Both are on the same chromosome.
- (any other valid point)

(Maximum 5 marks for differences)

	Point of comparison	Structural gene	Regulatory gene
1	Codes for?	Codes for a protein or RNA molecule that forms part of a structure or has an enzymatic function	Codes for a specific protein product that regulates the expression of the structural genes
2	Gene product interacts with DNA?	May not interact with DNA	Yes, e.g. the operator of the operon
3	Example?	<i>lacZ</i> = β -galactosidase gene <i>lacY</i> = permease gene <i>lacA</i> = transacetylase gene	<i>lacI</i> = <i>lac</i> repressor gene <i>trpR</i> = <i>trp</i> repressor gene gene for Catabolite Activator Protein (CAP)
4	Expression?	Regulated as in <i>lac</i> operon	Constitutive
5	Location?	Related structural genes found within operon	Regulatory genes exist singly outside operon (<i>lacI</i> is near <i>lac</i> operon) (gene for CAP is not anywhere near <i>lac</i> operon)
6	Presence of promoter/operator?	Both promoter and operator upstream of structural genes	Only promoter precedes it
7	Type of mRNA formed?	Polycistronic	Monocistronic

(b) Explain the roles of the operator and activator binding site in the *lac* operon.

[8]

(Any 8)

1. An operon is a unit of genetic function consisting of a promoter, an operator, and a coordinately regulated cluster of related (structural) genes whose products function in a common pathway.
2. Regulated / Controlled / Switched off and on / Transcribed together as a unit to produce a single messenger RNA (mRNA)
3. The operator can lie within the promoter or between the promoter and the structural genes. / The activator binding site can lie within or upstream of the promoter.
4. The operator is a binding site for the *lac* repressor.
5. Binding of *lac* repressor to the operator will deny the RNA polymerase access to the promoter and hence inhibit transcription.
6. Presence of the inducer allolactose / lactose inactivates the *lac* repressor by changing its conformation such that it can no longer bind to the operator, and transcription can be carried out / RNA polymerase can bind to the promoter.
7. The activator binding site is the binding site for Catabolite Activator Protein (CAP) / cAMP-CAP complex.
8. Low level of glucose leads to high level of cAMP in the cell, which results in high level of cAMP-CAP complex / activated CAP.
9. Binding of cAMP-CAP complex / activated CAP to the activator binding site will stimulate transcription / switch the *lac* operon on by increasing the affinity of RNA polymerase for the promoter.
10. Both the operator and activator binding site allow the expression of the lactose metabolizing enzymes to be responsive to changes in the environment (e.g. glucose and lactose concentrations) / prevent the waste of energy and resources.

- (c) Describe how the molecular structure of phospholipids is related to their function in the plasma membrane. [6]

(Any 6)

- Each phospholipid consists of a phosphate group, a glycerol backbone and two fatty acid chains.
- Each phospholipid is amphipathic / contains both a hydrophilic region and a hydrophobic region within the same molecule.
- Hydrophilic phosphate heads are on the outside of the bilayer, in contact with the surrounding aqueous medium.
- Hydrophobic fatty acid chains point towards the interior of the bilayer, away from the surrounding aqueous medium.
- Major component of the plasma membrane / Form a bilayer
- Selectively permeable to solutes due to presence of hydrophobic core in the bilayer
- Determine the fluidity of membrane
- The more unsaturated fatty acid chains are, the more fluid the membrane is.
- Kinks in unsaturated fatty acid chains prevent close packing of the phospholipids and decrease the interaction between adjacent fatty acid chains.
- Phospholipids with shorter fatty acid chains are more fluid.
- Shorter chain length reduces the tendency of the hydrocarbon tails to interact with one another.
- Some types of phospholipid can be split to produce products that function as second messengers in signal transduction.

[Total: 20]

- End of paper -